

Markov models specified in the thesis

Extracted from *Stochastic models of early Yersinia pestis infection: branching processes, bursting pathogenesis, and evolved strategy*

Working extraction

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Every entry gives the thesis location, state space, transition rates, parameters and attached conventions.
Rates are transcribed verbatim from the chapter sources.

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Scope and conventions

This document extracts every stochastic model in the thesis whose law is *specified* — that is, every model for which a state space and a transition mechanism (a transition matrix, a generator, a set of jump rates, or a transition kernel) are written down. Purely deterministic population models are included only where the thesis itself derives them as exact projections of a stochastic model, and this is flagged in the entry. Analysis carried out on a model (closed forms, quasi-stationarity, simulations) is not restated except where a derived object is part of the model’s working apparatus, such as the characteristic roots of the birth–death–catastrophe process or the flooding boundary.

House notation follows the thesis: $\mathbb{P}(\cdot)$, $\mathbb{E}(\cdot)$, $\text{Var}(\cdot)$; d roman; e roman; per-capita rates suppressed inside total rates by multiplication with the relevant count. Transition rates are written either as generator entries q_{ij} or as $\mathbb{P}(\Delta X = \cdot \mid \cdot) = \cdot \Delta t + o(\Delta t)$, exactly as in the source. Unless an entry says otherwise, time is continuous and the chain is time homogeneous; $o(\Delta t)$ terms are suppressed as in the sources.

The catalogue is organised by thesis chapter. Models restated in several chapters (the birth–death–catastrophe process above all) get one entry, whose source line lists every restatement. Model numbers (M1–M31) are used for cross-referencing within this document.

Chapter 1

Catalogue at a glance

	Model	State space	Where
M1	Simple random walk	$\mathbb{Z}$ (discrete time)	§2.2.1.1
M2	Galton–Watson process (general; binary case)	$\{0, 1, 2, \dots\}$ (discrete time)	§2.2.1.2; Ch. 3
M3	Poisson process	$\{0, 1, 2, \dots\}$	§2.2.2.2
M4	Yule process	$\{1, 2, \dots\}$	§2.2.2.3
M5	Linear birth–death process (incl. time-inhomogeneous rates)	$\{0, 1, 2, \dots\}$	§2.2.2.4; §2.2.3
M6	Birth–death–catastrophe process	$\{0, 1, 2, \dots\} \cup \{H\}$	§2.2.2.5; §4.2; §5.2.2
M7	Seasonal Lotka–Volterra process	$\{0, 1, \dots\}^2$	§2.2.3.1
M8	Logistic speciation process	$\{1, 2, \dots\}$, driven by an ODE	§2.2.3.2; §9.2
M9	Gated release (coupled ODE–CTMC)	$[0, \infty)^2 \times \{0, 1\}$	§2.2.4.1
M10	Surging gated release	$[0, \infty)^2 \times \{0, 1\}$	§2.2.4.2
M11	Absorption-only model	$\{0, \dots, N\}$ pairs	§2.B.1
M12	Absorption–death model	compartment pairs	§2.3.3.2
M13	Absorption–birth–death model	compartment pairs	§2.B.2
M14	Chained immediate transfer ($\mu = 0$ birth–catastrophe chain)	one load per cell, M cells	§5.10; §8.2.1
M15	Budding comparator cell	release/death event streams	§5.11; §6.2.2
M16	Basic Model of Viral Replication (deterministic projection)	$(\mathcal{I}, \mathcal{V})$ ODEs	§1.5; §6.2.1
M17	Burst-aware renewal system (kernel-driven population layer)	age-structured means	§6.3–6.4
M18	Branching approximation of early infection; flooding parameter L	offspring counts	§6.6
M19	Reset process (recurrent release)	$\{0, 1, 2, \dots\} + \text{cell death}$	§6.7.1
M20	Partial release (boolean thinning)	mean-field stock system	§6.7.2
M21	HIV extension (non-catastrophe kernels, sigmoid incidence)	kernel pair + incidence	§6.7.3
M22	Two-type birth–death–conversion with type-specific catastrophe	$\mathbb{Z}_{\geq 0}^2 \cup \{(R, R)\}$	§7.2
M23	Replicator–suppressor process	$\{0, \dots\} \times \{0, \dots, N\}$	§8.3.2
M24	Fixed-removal extension of the chained process	burst sizes + budget	§8.4
M25	Pimentel competitive-reversal chain	$\{0, \dots, N\} + \text{resistances}$	§9.3.2
M26	Fitness-coupled speciation variant	populations + fitness + species	§9.3.6
M27	Rise–run–ruin–rejuvenation (RRRR)	count + potential $(X_t, V(t))$	§9.4.2
M28	Multiplicative-dissipation automaton	$L \times L$ lattice + potentials	§9.5
M29	Birth–plague model (incl. vitality variant)	$\{0, 1, \dots\}^2$	§9.A

	Model	State space	Where
M30	Public-good accumulation process	count + good concentration	§9.B
M31	Quadrant argument (three- environment birth-death ordering)	three birth-death processes	§10.3.1

Chapter 2

Foundational processes (Chapter 2)

2.1 M1: Simple random walk

Specified in §2.2.1.1, equation (2.5).

State space. $\mathbb{Z}$, discrete time.

Transitions. Fix $q \in (0, 1)$:

$$P_{i,i+1} = q, \quad P_{i,i-1} = 1 - q. \quad (2.1)$$

Moments and law. $\mathbb{E}(X_n) = X_0 + n(2q - 1)$, $\text{Var}(X_n) = 4nq(1 - q)$; for deterministic X_0 , $\mathbb{P}(X_n = X_0 + 2r - n) = \binom{n}{r} q^r (1 - q)^{n-r}$, $r = 0, 1, \dots, n$.

Role. The elementary DTMC, and the embedded jump chain of the linear birth–death process (M5): the birth–death chain’s embedded walk has $q = \lambda/(\lambda + \mu)$. With absorbing barriers 0 and M , the gambler’s-ruin hitting probability is $h_i = (1 - \theta^i)/(1 - \theta^M)$ with $\theta = (1 - q)/q$, reducing to i/M at $q = \frac{1}{2}$.

2.2 M2: Galton–Watson branching process (general and binary)

Specified in §2.2.1.2, equations (2.9)–(2.12); developed throughout Chapter 3.

State space. $\{0, 1, 2, \dots\}$, discrete time (a Markov chain).

Transitions. With offspring law $\mathbb{P}(L = k) = p_k$ and mean $m = \mathbb{E}(L)$, from $Z_0 = 1$:

$$Z_{n+1} = \sum_{i=1}^{Z_n} L_i, \quad (2.2)$$

the L_i independent copies of L . Under the probability generating function $\phi(z) = \mathbb{E}(z^L)$ the generational PGFs compose: $\phi_{n+1} = \phi_n \circ \phi$, $\phi_0(z) = z$.

Binary (*simple*) case used throughout the thesis.

$$p_2 = p \text{ (division)}, \quad p_0 = 1 - p \text{ (death)}, \quad p_k = 0 \text{ } (k \notin \{0, 2\}), \quad (2.3)$$

with $m = 2p$, $\mathbb{E}(Z_n) = (2p)^n$, $\phi(z) = pz^2 + (1 - p)$, and the survival recursion

$$S_{n+1} = 2pS_n - pS_n^2, \quad S_0 = 1, \quad (2.4)$$

where $S_n = \mathbb{P}(Z_n > 0)$. For $0 < p \leq \frac{1}{2}$ survival decays to zero; for $p > \frac{1}{2}$ it decreases to the fixed point $(2p - 1)/p$.

Chapter 3 apparatus. For $0 < p < \frac{1}{2}$ the constant $A(p)$ is defined as the limit of $S_n/(2p)^n$ (existence via a convergent infinite product; degenerate endpoint $A(0) = \frac{1}{2}$); the population conditioned on survival settles to a quasi-stationary law with mean $1/A(p)$; near criticality $A(p) \sim 2(1 - 2p)$; rescaling the survival recursion to the logistic map identifies $A(p)$ with a value of the associated Koenigs linearising function.

2.3 M3: Poisson process

Specified in §2.2.2.2, equation (2.17).

State space. $\{0, 1, 2, \dots\}$; sole non-zero transition rate

$$q_{n,n+1} = \alpha, \quad n \geq 0. \quad (2.5)$$

Law. From $N_0 = 0$: $\mathbb{P}(N_t = n) = e^{-\alpha t}(\alpha t)^n/n!$, with $\mathbb{E}(N_t) = \text{Var}(N_t) = \alpha t$. Thinning with retention probability ϖ gives a Poisson process of rate $\varpi\alpha$.

2.4 M4: Yule process

Specified in §2.2.2.3, equation (2.19).

State space. $\{1, 2, \dots\}$ from a single founder; sole non-zero transition rate

$$q_{n,n+1} = \lambda n, \quad n \geq 1. \quad (2.6)$$

Law. $\mathbb{P}(Y_t = n) = e^{-\lambda t} \left(1 - e^{-\lambda t}\right)^{n-1}$, $\mathbb{E}(Y_t) = e^{\lambda t}$, $\text{Var}(Y_t) = e^{\lambda t} (e^{\lambda t} - 1)$; from k founders, the sum of k independent one-founder processes (negative binomial).

2.5 M5: Linear birth–death process, and time-inhomogeneous rates

Specified in §2.2.2.4, equation (2.22); §2.2.3 paragraph “Linear birth–death rates”.

State space. $\{0, 1, 2, \dots\}$; state 0 absorbing. Non-zero rates out of $n \geq 1$:

$$q_{n,n+1} = \lambda n, \quad q_{n,n-1} = \mu n. \quad (2.7)$$

$\mu = 0$ recovers the Yule process (M4).

Anchors. From $X_0 = N$ the mean is $Ne^{(\lambda-\mu)t}$. Extinction is certain for one founder when $\lambda \leq \mu$ and has probability μ/λ when $\lambda > \mu$; from N founders, $(\mu/\lambda)^N$. The embedded chain is M1 with $q = \lambda/(\lambda + \mu)$.

Time-inhomogeneous variant. Per-capita rates $\lambda(t), \mu(t)$. The mean closes: $\mathbb{E}(X_t | X_0 = n_0) = n_0 \exp\left(\int_0^t (\lambda(r) - \mu(r)) dr\right)$. The backward extinction probability $u(s, T) = \mathbb{P}(X_T = 0 | X_s = 1)$ satisfies the backward Riccati equation

$$-\frac{\partial u}{\partial s} = \mu(s) - (\lambda(s) + \mu(s))u + \lambda(s)u^2, \quad u(T, T) = 0, \quad (2.8)$$

whereas the forward-endpoint probability is $u_f(t) = B(t)/(1+B(t))$ with $B(t) = \int_0^t \mu(s) \exp\left(\int_0^s (\mu(r) - \lambda(r)) dr\right) ds$. The two endpoints must not be conflated.

2.6 M6: The birth–death–catastrophe (BDC) process

Specified in Definition in §2.2.2.5 (Definition 2.4); restated for the single-cell reading in §4.2 (Definition, equation (4.1)); canonical restatement in §5.2.2. Conventions: §4.2.3–4.2.4. Variants: several founders §5.9, chained transfer M14, killed convention below..

State space. $\{0, 1, 2, \dots\} \cup \{H\}$: the intracellular load while the cell (compartment) remains intact, with H a distinct absorbing rupture state. Both 0 and H are absorbing.

Transitions. Each replicative unit independently gives birth at rate λ , dies at rate μ , and triggers catastrophe (rupture) at rate δ . For $n \geq 1$ the non-zero transition rates are

$$q_{n,n+1} = \lambda n, \quad q_{n,n-1} = \mu n, \quad q_{n,H} = \delta n, \quad (2.9)$$

and, in first-step form,

$$\begin{aligned} \mathbb{P}(\Delta X_t = 1, \Delta W_t = 0 | X_t = n) &= \lambda n \Delta t + o(\Delta t) && \text{(birth)} \\ \mathbb{P}(\Delta X_t = -1, \Delta W_t = 0 | X_t = n) &= \mu n \Delta t + o(\Delta t) && \text{(death)} \\ \mathbb{P}(X_{t+\Delta t} = H, W_{t+\Delta t} = n | X_t = n) &= \delta n \Delta t + o(\Delta t) && \text{(catastrophe)} \\ \mathbb{P}(\Delta X_t = 0, \Delta W_t = 0 | X_t = n) &= 1 - (\lambda + \mu + \delta)n \Delta t + o(\Delta t) && \text{(no event)}. \end{aligned} \quad (2.10)$$

Default initial condition: $X_t = 1$ (working point $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$ in the Chapter 4 figures).

Release counter. With $\tau = \inf\{t : X_t = H\}$ the catastrophe time,

$$W_t = \begin{cases} 0, & t < \tau, \\ X_{\tau-}, & t \geq \tau, \end{cases} \quad \mathcal{K} := X_{\tau-} \quad (2.11)$$

is the released count and $\mathcal{K}$ the burst size. The left limit is essential.

Two conventions for rupture (§4.2.3). Sending rupture to 0 instead of H is *killing*; sending it to H is *catastrophe* (the thesis's convention). Killing makes internal extinction and rupture indistinguishable; with H separate one may distinguish the internal-extinction probability D , the rupture probability $1 - I$, and the productive probability $I_{\text{fix}} = I - D$.

Numerical-load convention (§4.2.4). Wherever X_t appears arithmetically, $H \mapsto 0$; $\mathcal{K}$ (burst size) is unrelated to $K(t) = \mathbb{E}(X_t^2)$ (second moment).

Characteristic roots and fixation functions (§5.2). With $\eta = (\lambda + \mu + \delta)/(2\lambda)$,

$$a = \eta + \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad b = \eta - \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad b < 1 < a, \quad (2.12)$$

$A := a - 1$, $B := 1 - b$, $\theta := \lambda(a - b) = \sqrt{(\lambda + \mu + \delta)^2 - 4\lambda\mu}$, and $AB = \delta/\lambda$. The no-burst and internal-extinction probabilities

$$I(t) = \mathbb{P}(\text{no burst by } t), \quad D(t) = \mathbb{P}(\text{internal extinction by } t, \text{ no burst}), \quad I_{\text{fix}}(t) = I(t) - D(t) \quad (2.13)$$

both satisfy the Riccati equation $\frac{dI}{dt} = \lambda(I - a)(I - b) = \mu - (\lambda + \mu + \delta)I + \lambda I^2$, differently initialised ($I(0) = 1$, $D(0) = 0$), with closed forms $I = \frac{aB + bAw}{B + Aw}$, $D = \frac{ab(w-1)}{aw-b}$, $I_{\text{fix}} = \frac{(a-b)^2 w}{(B + Aw)(aw-b)}$ at $w = e^{\theta t}$. b is the probability a cell clears internally without ever bursting. Anchors used later: $V_\infty = \mathbb{E}(W_t(\infty)) = \frac{a(1-b)}{a-1} = \frac{aB}{A}$, $\mathbb{E}(\mathcal{K} \mid \text{burst}) = a/(a-1)$, and the mean productive lifetime $\mathbb{E}[T_{\text{prod}}] = \lambda^{-1} \log \frac{a}{a-1}$.

Several founders (§5.9). From founding multiplicity k (MOI k), the fixation functions become $I(\alpha)^k - D(\alpha)^k$ with release kernel $g_k(\alpha) = \delta K_k(\alpha)$; families founded independently factor until the first catastrophe, which stops all lineages at once.

Provenance. The classical treatments cited in §5.2.2 (Brockwell; Karlin–Tregar; Di Crescenzo) allow immigration and distributed catastrophe sizes; the thesis specialises to the constant-rate, unit-catastrophe case above. A state-dependent catastrophe is not a mean-field hazard: $\mathbb{P}(\text{no catastrophe before generation } n) = \mathbb{E}\left((1 - \chi)^{\mathcal{E}_n}\right)$ over cumulative exposure $\mathcal{E}_n = \sum_{i < n} \tilde{X}_i$, and Jensen’s inequality forbids replacing $\mathcal{E}_n$ by its mean (§2.2.2.5, Remark).

2.7 M7: Seasonal Lotka–Volterra process

Specified in §2.2.3.1, equations (2.30)–(2.31).

State space. $\{0, 1, \dots\}^2$: prey X_t , predators Y_t .

Transitions. Time-inhomogeneous generator

$$q_{(x,y),(x+1,y)}(t) = b\mathcal{S}^+(t)x, \quad q_{(x,y),(x-1,y+1)}(t) = exy, \quad q_{(x,y),(x,y-1)}(t) = dy, \quad (2.14)$$

with seasonal multiplier $\mathcal{S}^+(t) = \max\{0, 1 + A \sin(\omega t)\}$ (clipped because a rate cannot be negative). For $A > 1$ each period $2\pi/\omega$ contains a closed season of length $(2/\omega) \arccos(1/A)$. On the predator-free axis the prey form a time-inhomogeneous Yule process; in the interior the first moments involve $\mathbb{E}(X_t Y_t)$ and do not close. The deterministic comparison system is $\dot{x} = b\mathcal{S}^+(t)x - exy$, $\dot{y} = exy - dy$. Exact paths are simulated by thinning against $\mathcal{S}^+(t) \leq 1 + A$.

2.8 M8: Logistic speciation process

Specified in §2.2.3.2, equations (2.33)–(2.37); full distribution in §9.2.

Driving environment. Total abundance $N(t)$ follows the logistic ODE $\frac{dN}{dt} = \varrho N \left(1 - \frac{N}{K}\right)$ (Chapter 9 notation; Chapter 2 uses r , C), with solution $N(t) = Ke^{\varrho t}/(\xi + e^{\varrho t})$, $\xi = K/N_0 - 1$.

State space and transitions. $S_t \in \{1, 2, \dots\}$, $S_0 = 1$, no extinction; a pure birth process with time-dependent per-capita rate

$$q_{s,s+1}(t) = \beta(t) s, \quad \beta(t) = \sigma(K - N(t)) = \sigma K \frac{\xi}{\xi + e^{qt}}. \quad (2.15)$$

Exactly a Yule process under a deterministic time change; one-way coupling (the species count does not feed back on N).

Law. With $\mathcal{B}(t) = \int_0^t \beta$, the marginal law is geometric, $\mathbb{P}(S_t = n) = e^{-\mathcal{B}(t)}(1 - e^{-\mathcal{B}(t)})^{n-1}$ for $n \geq 1$ (§9.2.4); $\mathbb{E}(S_t) = e^{\mathcal{B}(t)}$ converges to the finite limit $(K/N_0)^{\sigma K/\varrho}$ (Chapter 2: $(C/N_0)^{\sigma C/r}$), and $\mathcal{B}(\infty) = \frac{\sigma K}{\varrho} \ln \frac{K}{N_0}$, so S_∞ is a proper geometric limit.

Related numerical variant (§2.2.3.2). A logistic-environment birth–death comparison with origination $\beta_{\text{bd}}(t) = \sigma N'(t)$ and constant per-species extinction ε is used for figure-level comparison only, not as an interpretation of the model.

2.9 M9: Accumulation with gated release (coupled ODE–CTMC, two-way)

Specified in §2.2.4.1, equations (2.42)–(2.43).

State space. $(A_t, R_t, G_t) \in [0, \infty) \times [0, \infty) \times \{0, 1\}$: accumulated amount, transferred amount, gate. A piecewise-deterministic Markov process.

Flow (between gate jumps).

$$\frac{dA_t}{dt} = \rho - G_t c A_t, \quad \frac{dR_t}{dt} = G_t c A_t, \quad (2.16)$$

with external input $\rho > 0$ and transfer rate $c > 0$.

Jumps. The gate opens at a state-dependent rate and closes at a constant one:

$$\mathbb{P}(G_{t+\Delta t} = 1 \mid G_t = 0, A_t = a) = \eta a \Delta t + o(\Delta t), \quad \mathbb{P}(G_{t+\Delta t} = 0 \mid G_t = 1) = \gamma \Delta t + o(\Delta t). \quad (2.17)$$

Pathwise balance: $A_t + R_t = A_0 + R_0 + \rho t$. The moment $\frac{d}{dt}\mathbb{E}(A_t) = \rho - c\mathbb{E}(G_t A_t)$ is not closed. Illustration parameters: $\rho = 2$, $c = 4$, $\eta = 0.015$, $\gamma = 1.45$, $(A_0, R_0, G_0) = (5, 0, 0)$.

2.10 M10: Accumulation with surging gated release (coupled ODE–CTMC, mode-driven)

Specified in §2.2.4.2, equation (2.44).

Specification. The flow and closing rate of M9 are retained; opening is autonomous,

$$\mathbb{P}(G_{t+\Delta t} = 1 \mid G_t = 0) = \nu \Delta t + o(\Delta t), \quad (2.18)$$

so G_t is an autonomous two-state CTMC and the model sits in case (i) of the coupled taxonomy. The gate mean closes: $\mathbb{E}(G_t) \rightarrow \nu/(\nu + \gamma)$; the balance law is unchanged. Illustration parameters: as M9 with $\nu = 0.3$.

Context. §2.2.4 also states the three-way taxonomy of couplings (chain switching the vector field; prescribed trajectory driving the rates; two-way) with joint generator $(\mathcal{L}f)(\mathbf{y}, i) = F(\mathbf{y}, i) \cdot \nabla_{\mathbf{y}} f + \sum_{j \neq i} q_{ij}(\mathbf{y})[f(\mathbf{y}, j) - f(\mathbf{y}, i)]$; M9 is the two-way case and M10 the first case.

2.11 M11: Absorption-only model

Specified in Appendix 2.B, §2.B.1, transition list after (2.48).

Setting. One compartment drawing particles from a medium; X_t particles inside, Y_t outside; $X_0 = 0$, $Y_0 = N$.

Transitions.

$$\mathbb{P}(\Delta X = 1, \Delta Y = -1 \mid Y_t = j) = j\alpha \Delta t \quad (\text{absorption}) \quad (2.19)$$

and no event otherwise. Particles are neither created nor destroyed ($p_{i,j}(t) = 0$ unless $i + j = N$).

Law. $X_t \sim \text{Binomial}(N, 1 - e^{-\alpha t})$; single-particle PGF $G_1 = (1 - e^{-\alpha t})x + e^{-\alpha t}y$.

2.12 M12: Absorption–death model

Specified in §2.3.3.2 worked example, transition list before (2.52).

Transitions. To the absorption channel of M11 add interior death:

$$\begin{aligned} \mathbb{P}(\Delta X = 1, \Delta Y = -1 \mid Y_t = j) &= j\alpha \Delta t, & (\text{absorption}) \\ \mathbb{P}(\Delta X = -1, \Delta Y = 0 \mid X_t = i) &= i\mu \Delta t, & (\text{death}) \end{aligned}$$

with $X_0 = 0$, $Y_0 = N$. Single-particle state probabilities: $p_{0,1}(t) = e^{-\alpha t}$, $p_{1,0}(t) = \frac{\alpha}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})$, $p_{0,0} = 1 - p_{0,1} - p_{1,0}$; the N -particle PGF is the N th power of the single-particle one (equation (2.54) of the thesis).

Role. The worked example for the method of characteristics; the generator-level PDE is $\frac{\partial G}{\partial t} = \mu(1 - x)\frac{\partial G}{\partial x} + \alpha(x - y)\frac{\partial G}{\partial y}$.

2.13 M13: Absorption–birth–death model

Specified in Appendix 2.B, §2.B.2–2.B.3, transitions before (2.58); closed form Proposition 2.B.1.

Transitions. Absorption as M11/M12 plus interior birth:

$$\begin{aligned} \mathbb{P}(\Delta X = 1, \Delta Y = -1 \mid Y_t = j) &= j\alpha \Delta t, & (\text{absorption}) \\ \mathbb{P}(\Delta X = 1, \Delta Y = 0 \mid X_t = i) &= i\lambda \Delta t, & (\text{birth}) \\ \mathbb{P}(\Delta X = -1, \Delta Y = 0 \mid X_t = i) &= i\mu \Delta t, & (\text{death}) \end{aligned}$$

with $X_0 = 0$, $Y_0 = N$.

Closed form. With $x_+ = 1$, $x_- = \mu/\lambda$, $b_1 = \lambda - \mu$, $\omega = \alpha/b_1$, $W = (x - x_+)/ (x - x_-)$ and $\Psi(\zeta) = x_+ + \frac{\alpha}{\lambda} \sum_{n \geq 1} \frac{\zeta^n}{n + \omega} = x_+ + (x_+ - x_-)({}_2F_1(1, \omega; \omega + 1; \zeta) - 1)$,

$$G(x, y, t) = \left\{ e^{-\alpha t} [y - \Psi(W)] + \Psi(W e^{b_1 t}) \right\}^N. \quad (2.20)$$

The parameter-regular convolution representation is $g(x, y, t) = e^{-\alpha t} y + \alpha e^{-\alpha t} \int_0^t e^{\alpha v} F(x, v) dv$ with F the interior birth–death PGF, and $G = g^N$. Resonance at $\alpha = n(\mu - \lambda)$, $n = 1, 2, \dots$; the generating function stays regular there.

Note. These three absorption models share the compartment geometry of Figure 2.5 of the thesis; nothing returns from the compartment to the medium. They are a family solved for method rather than used by the modelling chapters (Chapter 10, contributions table).

Chapter 3

The thesis process and its single-cell comparators (Chapters 4–5)

The birth–death–catastrophe process is M6; Chapters 4 and 5 restate it (canonical notation table in §5.2.1) and develop its distribution theory. The two further specifications in this block are the chained-transfer laboratory and the budding comparator.

3.1 M14: Chained immediate transfer ($\mu = 0$ birth–catastrophe chain)

Specified in §5.10 (§5.10.1–5.10.2); described as “successive birth–death–catastrophe” in §8.2.1.

Specification. M host cells, all empty but one holding a single particle. Each runs the $\mu = 0$ BDC process (M6 with $\mu = 0$, so only births λn and catastrophe δn); on rupture the entire released load passes immediately and wholesale into a fresh cell, founding the next process with that load as initial census. No extracellular phase, no clearance, no re-infection; all M ruptures occur almost surely.

Anchors. With $s = \delta/(\lambda + \delta)$ and $\rho = \lambda/(\lambda + \delta) = 1 - s$: event types ignore the load (birth with probability ρ , catastrophe with probability s), the births before catastrophe in a cell founded with any load are $G \sim \text{Geom}_0(s)$, and the k th rupture size is $r_k = 1 + G_1 + \cdots + G_k$ with

$$\mathbb{P}(r_k = n) = \binom{n+k-2}{k-1} s^k \rho^{n-1}, \quad n \geq 1. \quad (3.1)$$

The case $\mu > 0$ is recorded as open.

3.2 M15: Budding comparator cell

Specified in §5.11, equations (5.28)–(5.29); restated §6.2.2, equations (6.4)–(6.5).

Specification. A budding infected cell experiences two independent Poisson event streams: release of one particle at rate p and death of the cell at rate $d_{\mathcal{I}}$, continuing until death.

Anchors.

$$S_{\text{bud}}(\alpha) = e^{-d_I \alpha}, \quad V_{\text{bud}}(\alpha) = \frac{p}{d_I} \left(1 - e^{-d_I \alpha}\right), \quad (3.2)$$

and the total number released is geometric on $\{0, 1, 2, \dots\}$:

$$\mathbb{P}(\mathcal{K}_{\text{bud}} = k) = \frac{d_I}{d_I + p} \left(\frac{p}{d_I + p}\right)^k, \quad k = 0, 1, 2, \dots \quad (3.3)$$

Structural contrast with bursting: in budding, release and death are independent events; in bursting (M6) they are the same event.

Chapter 4

The population layer (Chapter 6)

4.1 M16: Basic Model of Viral Replication (BMVR)

Specified in §6.2.1, equations (6.1)–(6.2); three-companion version in §1.5.

Specification (target-cell-constant). Infected cells $\mathcal{I}$ and free particles $\mathcal{V}$ obey

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{d\mathcal{V}}{dt} = p \mathcal{I} - c \mathcal{V}, \quad R_0 = \frac{\gamma T p}{c d_{\mathcal{I}}}, \quad (4.1)$$

with target cells T fixed, infection rate γ , virion clearance c , release rate p , removal rate $d_{\mathcal{I}}$. *Deterministic*; the thesis shows it is the exact one-cell limit of the budding cell M15 and a projection of the renewal system M17, not a Markov model itself. §1.5 gives the full three-companion version $\frac{dT}{dt} = s - \gamma TV$, $\frac{dI}{dt} = \gamma TV - d_I I$, $\frac{dV}{dt} = pI - cV$.

4.2 M17: The burst-aware renewal system

Specified in §6.3 (kernels, equations (6.7)–(6.9)) and §6.4.1 (system, Definition 6.4.1); exactness Proposition 6.4.2; skeleton §6.4.3.

Intracellular layer. The load inside a cell of age α follows M6. Two kernels:

$$S(\alpha) := I_{\text{fix}}(\alpha) = \frac{(a-b)^2 e^{\theta\alpha}}{(B + Ae^{\theta\alpha})(ae^{\theta\alpha} - b)}, \quad g(\alpha) := \delta K(\alpha) = V'(\alpha), \quad (4.2)$$

the survival kernel (probability still productively infected at age α) and the release kernel (expected release flux at age α); g is explicit because a burst fires at rate δX_α and releases X_α , giving $\delta \mathbb{E}[X_\alpha^2] d\alpha$.

Population system. With incidence $i(t) = \gamma T \mathcal{V}(t)$ and $\mathcal{I}_0$ initially infected cells of age zero:

$$\mathcal{I}(t) = \mathcal{I}_0 I_{\text{fix}}(t) + \int_0^t i(t-\alpha) I_{\text{fix}}(\alpha) d\alpha = \mathcal{I}_0 I_{\text{fix}} + (i * I_{\text{fix}})(t), \quad (4.3)$$

$$\frac{d\mathcal{V}}{dt}(t) = \mathcal{I}_0 g(t) + \int_0^t i(t-\alpha) g(\alpha) d\alpha - c \mathcal{V}(t) = \mathcal{I}_0 g + (i * g)(t) - c \mathcal{V}(t). \quad (4.4)$$

These are the first moments *exactly* of the stochastic process in which each infection starts an independent BDC cell and each particle infects at rate γT or clears at rate c (linear incidence is what closes the hierarchy; assumption (A1) of §6.1.1).

Skeleton. Any kernel pair (S, g) induces

$$\mathcal{I}(t) = \mathcal{I}_0 S(t) + (i * S)(t), \quad \frac{d\mathcal{V}}{dt} = \mathcal{I}_0 g(t) + (i * g)(t) - c \mathcal{V}(t), \quad (4.5)$$

with effective parameters $p_{\text{eff}}(r) = \tilde{g}(r)/\tilde{S}(r)$ and, for linear incidence, $R_0 = (\gamma T/c) \int_0^\infty g$. Standing assumptions (A1)–(A5): constant T ; continuum convention for infection (virions not consumed); synchronous zero-age founding cohort at $t = 0$; independent cells, single infection; no spatial structure.

4.3 M18: Branching approximation of the early phase; the flooding parameter

Specified in §6.6.1–6.6.2, equations (6.27)–(6.33).

Specification. Each released particle independently infects a target cell at rate γT or is cleared at rate c , hence infects with probability

$$q = \frac{\gamma T}{\gamma T + c}. \quad (4.6)$$

The offspring of one infected cell is the burst size $\mathcal{K}$ (M6) thinned by q : offspring PGF

$$G_{\text{off}}(z) = b + \frac{\delta}{\lambda} \frac{y}{a - y}, \quad y = 1 - q + qz, \quad (4.7)$$

including the mass b of cells that never burst. Mean $m = qV_\infty$ ($\approx R_0$ when $\gamma T \ll c$). The comparator (M15) is matched at equal total output $p/d_{\mathcal{I}} = V_\infty$; its thinned offspring law is geometric with the same mean.

Flooding parameter and criterion.

$$L := a(1 - b) = a - \frac{\mu}{\lambda}, \quad \text{so that} \quad V_\infty = \frac{L}{a - 1}. \quad (4.8)$$

For $m > 1$ the establishment extinction probabilities satisfy $z_{\text{ext}}^{\text{burst}} - z_{\text{ext}}^{\text{bud}} = (L - 1) \left(\frac{1}{m} - 1 \right)$, so bursting is advantageous if and only if

$$\delta(\lambda + \mu) > \lambda\mu \quad \Longleftrightarrow \quad \frac{1}{\delta} < \frac{1}{\lambda} + \frac{1}{\mu} \quad (4.9)$$

(with the condition holding for all $\delta > 0$ when $\mu = 0$).

4.4 M19: Reset process (recurrent release)

Specified in §6.7.1, equations (6.41)–(6.43).

Specification. While the cell is alive, the load undergoes birth at rate λn , death at rate μn , optional immigration at rate ν , and a *dump that resets*: at rate δn the whole pool is released and the load returns to zero, after which it may re-accumulate; the cell dies at a separate, load-independent rate $d_{\mathcal{I}}$. Dumps do not absorb.

Kernels and anchors.

$$S(\alpha) = e^{-d_I \alpha}, \quad g(\alpha) = \delta e^{-d_I \alpha} \mathbb{E}[X_\alpha^2], \quad p_{\text{eff}}(r) = (r + d_I) \tilde{g}(r), \quad d_{I,\text{eff}} \equiv d_I, \quad (4.10)$$

with $\mathbb{E}[X_\alpha^2]$ computed for the non-absorbing reset process. $R_0 = (\gamma T/c) \int_0^\infty g$ is finite only for $d_I > 0$; in the mature-cell limit, $p_\infty = \delta \mathbb{E}_\pi[X^2]$ for stationary law π .

4.5 M20: Partial release (boolean thinning)

Specified in §6.7.2, equation (6.44).

Specification. Release events fire on the load-dependent clock δX and remove each unit independently with probability φ (the default); at mean-field level, with Q the total intracellular stock across productive cells,

$$\frac{dI}{dt} = \gamma T \mathcal{V} - d_I I, \quad \frac{dQ}{dt} = \nu I - \delta \varphi \frac{Q^2}{I} - d_I Q, \quad \frac{d\mathcal{V}}{dt} = \delta \varphi \frac{Q^2}{I} - c \mathcal{V}. \quad (4.11)$$

With a load-independent clock $\lambda(X) = \delta$ instead, mean export is $\delta \varphi Q$ and the equations collapse to stock-and-export dynamics with $\varepsilon_{\text{eff}} = \delta \varphi$: partial release is invisible in the mean field exactly when the clock is load-independent. Antiviral mapping: blocking production lowers λ ; enhancing degradation raises μ ; blocking entry lowers γT ; blocking secretion reduces φ (with $\varphi = 1 - \epsilon_s$).

4.6 M21: HIV extension

Specified in §6.7.3, equation (6.45).

Specification. HIV is not carried by the catastrophe process: the mean load follows an immigration-like $\bar{x}' \approx \nu - \mu \bar{x}$. The skeleton of M17 carries HIV on a different kernel pair $(S_{\text{HIV}}, g_{\text{HIV}})$ — probability of still being in a productive lineage, and density of productive sojourn times — with a multi-stage eclipse (near-zero release for small α) supplying the delay; the effective-parameter map carries over verbatim. Near the observed establishment threshold the data favour a sigmoid incidence, an Allee effect:

$$f(\mathcal{V}, T) = \gamma T \mathcal{V} \cdot \frac{\mathcal{V}}{K_A + \mathcal{V}}, \quad (4.12)$$

under which the first-moment exactness of M17 no longer holds.

Chapter 5

Two-type extension (Chapter 7)

5.1 M22: Two-type birth–death–conversion with type-specific catastrophe rates

Specified in §7.2.1 (Definition and rate list), §7.2.2 (occupation-time representation), §7.2.4 (dimensionless form); biological reading in §7.5.

State space. $(x, y) \in \mathbb{Z}_{\geq 0}^2$ before absorption; a single global absorbing state (R, R) .

Transitions. Type 1 gives birth at rate λ_1 , dies at rate μ_1 , and converts irreversibly to type 2 at rate ν ; type 2 gives birth at rate λ_2 and dies at rate μ_2 ; all rates non-negative. Before absorption, the non-zero transition rates from (x, y) are

$$\begin{array}{ll}
 (x, y) \longrightarrow (x + 1, y) & \text{at rate } \lambda_1 x, \\
 (x, y) \longrightarrow (x - 1, y) & \text{at rate } \mu_1 x, \\
 (x, y) \longrightarrow (x - 1, y + 1) & \text{at rate } \nu x, \\
 (x, y) \longrightarrow (x, y + 1) & \text{at rate } \lambda_2 y, \\
 (x, y) \longrightarrow (x, y - 1) & \text{at rate } \mu_2 y, \\
 (x, y) \longrightarrow (R, R) & \text{at rate } \delta_1 x + \delta_2 y,
 \end{array} \tag{5.1}$$

the last being the catastrophe: one individual contributes a type-specific rate, and the first trigger absorbs the whole process.

Object of study. $S(t) = \mathbb{P}_{(1,0)}\{\tau_c > t\}$ and $G(t) = \mathbb{P}_{(0,1)}\{\tau_c > t\}$: no-catastrophe probabilities, not population-survival probabilities. They admit the occupation-time representation

$$S(t) = \mathbb{E}_{(1,0)} \left[\exp \left\{ - \int_0^t (\delta_1 X_s + \delta_2 Y_s) ds \right\} \right], \quad G(t) = \mathbb{E}_{(0,1)} \left[\exp \left\{ - \int_0^t (\delta_1 X_s + \delta_2 Y_s) ds \right\} \right], \tag{5.2}$$

and satisfy the triangular backward system

$$\frac{dS}{dt} = \lambda_1 S^2 - (\lambda_1 + \mu_1 + \nu + \delta_1)S + \mu_1 + \nu G, \quad S(0) = 1, \quad S'(0) = -\delta_1, \tag{5.3}$$

$$\frac{dG}{dt} = \lambda_2 G^2 - (\lambda_2 + \mu_2 + \delta_2)G + \mu_2, \quad G(0) = 1, \quad G'(0) = -\delta_2. \tag{5.4}$$

Dimensionless form: time in units of λ_1 ; rates $\hat{\mu}_i = \mu_i/\lambda_1$, $\hat{\nu} = \nu/\lambda_1$, $\hat{\delta}_i = \delta_i/\lambda_1$, $\hat{\lambda}_2 = \lambda_2/\lambda_1$. The main formula assumes $\hat{\lambda}_2 > 0$, $\hat{\delta}_2 > 0$, $\hat{\nu} > 0$; the zero-birth and zero-catastrophe boundaries are treated separately, and with the appropriate initial data the zero-catastrophe case recovers the two-type branching equations of Antal and Krapivsky.

Biological reading (§7.5). The types may describe early and adapted intracellular phenotypes of *Y. pestis*, with catastrophe representing irreversible loss of macrophage containment; four catastrophe-rate regimes (per-type constant, type-2 only, and mixtures) are compared exactly.

Chapter 6

Pathogenesis: depletion models (Chapter 8)

6.1 M23: Replicator–suppressor process

Specified in §8.3.2, Definition (equations (8.2)–(8.3)); first-step analysis §8.3.3; fates §8.3.4.

Setting. A leucocyte has engulfed pathogens and holds a finite, depletable store of toxic granules; granules are consumed as they kill.

State space. $\mathcal{S} = \{(i, j) : i \in \mathbb{N} \cup \{0\}, j \in \{0, \dots, N\}\}$: X_t infectious agents in one host cell, Q_t remaining granules (store size N fixed at engulfment, never replenished).

Transitions. For $(i, j) \in \mathcal{S}$:

$$\mathbb{P}(\Delta X_t = 1, \Delta Q_t = 0 \mid X_t = i, Q_t = j) = \lambda i \Delta t + o(\Delta t) \quad (\text{birth}), \quad (6.1)$$

$$\mathbb{P}(\Delta X_t = -1, \Delta Q_t = -1 \mid X_t = i, Q_t = j) = \varsigma i j \Delta t + o(\Delta t) \quad (\text{suppression}), \quad (6.2)$$

$$\mathbb{P}(\Delta X_t = 0, \Delta Q_t = 0 \mid X_t = i, Q_t = j) = 1 - (\lambda i + \varsigma i j) \Delta t + o(\Delta t) \quad (\text{no event}). \quad (6.3)$$

The product $\varsigma i j$ is the only nonlinearity. The column $X_t = 0$ is extinction; the row $Q_t = 0$ is a spent store, from which the process is a pure birth process and survival is certain. Interest centres on $\mathbb{P}(\text{infection survives}) = 1 - \lim_{t \rightarrow \infty} \mathbb{P}(X_t = 0)$; survival is certain whenever the founding cohort outnumbers the store ($i > j$), and in the contested region $1 \leq i \leq j$ the survival map $\pi_{i,j}$ is computed by first-step analysis.

6.2 M24: Fixed-removal extension of the chained process

Specified in §8.4.1–8.4.2, equations (8.5), (8.8), (8.9); assumptions in Remarks 8.4.1–8.4.2.

Specification. The chained-rupture picture (M14’s complete transfer, or dispersal to new target cells) is extended by one assumption: after each burst release, a fixed count of agents is removed. The removal rule is

$$\text{removal} = \begin{cases} r, & r \leq \vartheta, \\ \vartheta, & r > \vartheta, \end{cases} \quad \text{that is, } \min(r, \vartheta), \quad (6.4)$$

with r the burst size and ϑ the fixed removal budget (an idealisation of a saturating casualty law $\vartheta(r) \sim \log(r + 1)$). Under the framing used for the calculation, each surviving released agent founds a new infection in a new target cell.

Assumptions. Killer-cell prevalence resets between rupture events (steady state, uniform distribution — this makes ϑ the same at every delivery and the independence exact); the burst-size distribution is that of M6 including internal-death outcomes ($\mathbb{E}(r) = V_\infty$).

Quantity of interest. The expected replication ratio

$$\mathcal{R}(\vartheta) = \mathbb{E}(\max(0, r - \vartheta)) = [\mathbb{E}(r \mid r \geq \vartheta) - \vartheta] \mathbb{P}(r \geq \vartheta), \quad (6.5)$$

the expected number of newly infected target cells created by a host founded from one initial bacterium. Deliberately not the R_0 of Chapter 6; the relation is stated in §8.5.1.

Chapter 7

Models with non-constant rates (Chapters 9–10)

7.1 M25: Pimentel competitive-reversal chain

Specified in §9.3.2, equations (9.6)–(9.9); critical point §9.3.4.

State space. $A_t \in \{0, 1, \dots, N\}$ in discrete time, two populations with $A_t + B_t = N$; $i = 0$ and $i = N$ absorbing. The triple $(A_t, R_{a,t}, R_{b,t})$ is Markov; A_t alone is not.

Transition kernel. For $0 < i < N$:

$$\mathbb{P}(A_{t+1} = j \mid A_t = i) = \begin{cases} q_a = \frac{i}{N}(1 - p_b), & j = i + 1, \\ q_b = \left(1 - \frac{i}{N}\right)(1 - p_a), & j = i - 1, \\ q_c = \frac{i}{N}p_b + \left(1 - \frac{i}{N}\right)p_a, & j = i, \\ 0 & \text{otherwise.} \end{cases} \quad (7.1)$$

Blocking capacities

$$p_a = \max \left\{ 0, 1 - \frac{R_b}{R_a} \right\}, \quad p_b = \max \left\{ 0, 1 - \frac{R_a}{R_b} \right\}, \quad (7.2)$$

so at most one of p_a, p_b is non-zero. Resistances are driven up when the population is in a minority:

$$R_{k,t+1} = R_{k,t} + \varepsilon \Delta R_{k,t}, \quad \Delta R_{a,t} = \max \left\{ 0, \left(\frac{N - A_t}{A_t} \right)^\alpha \right\}, \quad \Delta R_{b,t} = \max \left\{ 0, \left(\frac{A_t}{N - A_t} \right)^\alpha \right\}, \quad (7.3)$$

with $R_{a,0} = R_{b,0} = 1$ without loss (only ε/R_0 matters). The critical state $q_a = q_b$ is $i_c = N^{\frac{1-p_a}{2-p_a}}$ ($p_a > 0$) or $N^{\frac{1}{2-p_b}}$ ($p_b > 0$), and is unstable; it is a function of the path, not the parameters alone.

7.2 M26: Fitness-coupled speciation variant (“Pimentel plus”)

Specified in §9.3.6, equations (9.22)–(9.25); recorded as specified but never simulated.

Specification. The kernel of M25 recast with fitness in place of resistance, a species count carried alongside and acting back on fitness:

$$\mathbb{P}(\Delta A_t = 1) = \frac{A_t}{N} \frac{f_a}{\bar{f}}, \quad \mathbb{P}(\Delta A_t = -1) = \left(1 - \frac{A_t}{N}\right) \frac{f_b}{\bar{f}}, \quad (7.4)$$

with $\bar{f}$ the mean fitness across the domain; fitness updates

$$f_a(t + \Delta t) = f_a(t) + \varepsilon \left(\frac{B_t}{A_t}\right)^\alpha - \gamma S_t^{(a)}, \quad f_b(t + \Delta t) = f_b(t) + \varepsilon \left(\frac{A_t}{B_t}\right)^\alpha - \gamma S_t^{(b)}, \quad (7.5)$$

and per-population species counts evolving as birth–death processes with constant extinction μ_a and dominance-and-fitness speciation

$$\mathbb{P}(\Delta S_t^{(a)} = 1) = \beta_a S_t^{(a)} \Delta t, \quad \beta_a = \left(\frac{A_t}{N}\right)^\zeta f_a. \quad (7.6)$$

The penalty $-\gamma S$ is the endogenous weakening of the winner; ζ controls how firmly dominance must be established before variation is released.

7.3 M27: Rise–run–ruin–rejuvenation (RRRR)

Specified in §9.4.2, equations (9.26)–(9.27).

State space. $X_t \in \{0, 1, 2, \dots\}$ replicative agencies with an auxiliary real potential $V(t)$; the pair $(X_t, V(t))$ is Markov, X_t alone is not. Described in the source as “a birth–death–catastrophe process with catastrophe rate $\delta X_t/V(t)$ and an auxiliary ODE for V ”.

Specification. Potential flows in and is consumed:

$$\frac{dV}{dt} = \phi - c X_t. \quad (7.7)$$

Given V , the count moves by

$$\begin{aligned} \mathbb{P}(\Delta X_t = 1 \mid X_t) &= \lambda X_t \Delta t + o(\Delta t) && \text{(proliferation),} \\ \mathbb{P}(X_{t+\Delta t} = 0 \mid X_t) &= \delta \frac{X_t}{V(t)} \Delta t + o(\Delta t) && \text{(catastrophe),} \\ \mathbb{P}(X_{t+\Delta t} = 1 \mid X_t = 0) &= \omega \Delta t + o(\Delta t) && \text{(revival),} \\ \mathbb{P}(X_{t+\Delta t} = X_t) &= 1 - \left(\lambda X_t + \delta \frac{X_t}{V(t)} + \omega \mathbf{1}\{X_t = 0\} \right) \Delta t + o(\Delta t) && \text{(no event).} \end{aligned} \quad (7.8)$$

Catastrophe is driven by per-capita scarcity; revival keeps 0 from absorbing, making the process recurrent. V stays positive for all time (the hazard diverges as $V \downarrow 0$). Illustration parameters: $\phi = 4$, $c = 0.8$, $\lambda = 0.5$, $\delta = 0.25$, $\omega = 0.15$.

7.4 M28: Multiplicative-dissipation automaton (speciation as catastrophe)

Specified in §9.5 (§9.5.2 state, §9.5.3 competition, §9.5.4 mutation and potential, §9.5.5 fragmentation).

State space. An $L \times L$ lattice ($L = 256$; Moore neighbourhood, periodic boundary, torus), each cell empty or occupied. Every occupied cell carries a species and, within it, a sub-species. For each extant species i : accumulated potential V_i (real), occupied cells N_i , extant sub-species $X_i \leq N_i$. Continuous updates (potential, mutation) advance in chunks between discrete contests (one attempt per site per sweep).

Competition. A cell ζ carrying sub-species u of species i has strength

$$w(\zeta) = V_i \times (n_\zeta + 1)^\theta \times e^{-\nu|u|/\kappa_i}, \quad \kappa_i = \frac{N_i}{X_i}, \quad (7.9)$$

with n_ζ matching neighbours at the contest level and $|u|$ the area of u ; the contest goes to ζ with probability $w(\zeta)/(w(\zeta) + w(\zeta'))$. Between sub-species of one species the potential cancels; between distinct species the rarity factor is omitted, $\mathbb{P}(A \text{ wins}) = \frac{V_A(n_A+1)^\theta}{V_A(n_A+1)^\theta + V_B(n_B+1)^\theta}$. Contests against empty ground: fixed strength d , an occupied cell takes a contested empty neighbour with probability $1 - d$, empty ground never invades.

Mutation and potential. Sub-species arise within a species at species-wide rate $R_i^{\text{mut}} = m V_i N_i$ (per-cell rate $m V_i$). Each occupied cell supplies potential at rate φ and each extant sub-species costs c :

$$\frac{dV_i}{dt} = \varphi N_i - c X_i, \quad \text{floored at } 0, \quad (7.10)$$

so potential can fall at all only if $c > \varphi$.

Fragmentation. When a species carries too many sub-species it fragments: every sub-species becomes an independent species at once (nothing dies; the interaction law changes). Two triggers, tested in order: deterministically if the potential update would take V_i to zero or below; otherwise with hazard

$$R_i^{\text{frag}} = \delta \frac{X_i}{V_i}, \quad (7.11)$$

M27's catastrophe rate with the count of individuals replaced by the count of types. No revival term is needed: the catastrophe restarts the process.

7.5 M29: Birth–plague model, and the vitality-coupled variant

Specified in Appendix 9.A, equations (9.A.1)–(9.A.3); simulated but not analysed.

State space. X_t uninfected individuals, Y_t infected.

Transitions.

$$\mathbb{P}(\Delta X_t = 1, \Delta Y_t = 0 \mid X_t, Y_t) = \lambda X_t \Delta t + o(\Delta t) \quad (\text{birth}), \quad (7.12)$$

$$\mathbb{P}(\Delta X_t = -1, \Delta Y_t = 1 \mid X_t, Y_t) = (\delta + \chi Y_t) X_t \Delta t + o(\Delta t) \quad (\text{corruption/infection}), \quad (7.13)$$

$$\mathbb{P}(\Delta X_t = 0, \Delta Y_t = -1 \mid X_t, Y_t) = \mu Y_t \Delta t + o(\Delta t) \quad (\text{clearance}). \quad (7.14)$$

δ is the spontaneous-corruption rate (linear in X_t) and χ the contact-infection coefficient (bilinear $X_t Y_t$); the two mechanisms superpose in one channel. Illustration parameters: $(\lambda, \delta, \chi, \mu) = (1, 0.1, 0.02, 10)$.

Vitality-coupled variant. Carry $V(t)$ with $\frac{dV}{dt} = \phi - cX_t$ and replace the middle channel by

$$\mathbb{P}(\Delta X_t = -1, \Delta Y_t = 1 \mid X_t, Y_t) = \left(\frac{\delta}{V(t)} + \chi Y_t \right) X_t \Delta t + o(\Delta t), \quad (7.15)$$

the other channels unchanged; a population that grows large drives V down and raises its own loss rate. Not simulated.

7.6 M30: Public-good accumulation process

Specified in Appendix 9.B, equation (9.B.1); minimum founding cohort (9.B.2).

State space. X_t *Y. pestis* within one macrophage and $P(t)$ the public-good concentration.

Specification.

$$\begin{aligned} \frac{dP}{dt} &= gX_t - c, \\ \mathbb{P}(\Delta X_t = 1 \mid X_t) &= \lambda(t) X_t \Delta t + o(\Delta t), \quad \lambda(t) = \lambda_0 + \alpha P(t), \\ \mathbb{P}(\Delta X_t = -1 \mid X_t) &= \mu X_t \Delta t + o(\Delta t), \end{aligned} \quad (7.16)$$

with production rate g per bacterium, clearance c , base birth rate λ_0 , public-good advantage α , death rate μ (host toxicity folded in). The regime of interest is $\mu > \lambda_0$: subcritical alone, supercritical once enough good has accumulated.

Embedded Galton–Watson reading. In a host-to-host sequence the infection is a Galton–Watson process, supercritical exactly when $\lim_{t \rightarrow \infty} \mathbb{P}(X_t > 0 \mid X_0 = K) > \frac{1}{2}$; the minimum founding cohort is

$$K^* = \inf \left\{ K : \lim_{t \rightarrow \infty} \mathbb{P}(X_t > 0 \mid X_0 = K) > \frac{1}{2} \right\}. \quad (7.17)$$

Illustration parameters: $(\lambda_0, \alpha, g, c) = (0.6, 0.25, 0.15, 0.45)$, so P falls whenever $X_t < c/g = 3$.

7.7 M31: The quadrant argument (three-environment birth–death ordering)

Specified in §10.3.1, equations (10.4)–(10.6).

Specification. Replication in each of three environments — inside a macrophage (M), inside a neutrophil (N), extracellular (E) — treated as a simple birth–death process (M5) with rates (β_i, μ_i) , $i \in \{M, N, E\}$, with

$$\mu_N > \mu_E > \mu_M \quad \text{and} \quad \beta_M = \beta_E = \beta_N = \beta, \quad (7.18)$$

so the net rates order as $\beta_M - \mu_M > \beta_E - \mu_E > \beta_N - \mu_N$. A consistency argument, not a derivation: the ordering is inferred from the strategy it is meant to explain. (The surrounding discussion notes that with depletable granules, as in M23, μ_M and μ_N are in reality declining functions of the cohort’s history.)

Where each model sits in the thesis argument

- **M1–M5, M7–M13** (Chapter 2) are the toolbox: each is developed once, on whichever process displays a method most cleanly. M6 is introduced there and becomes the thesis’s central object.
- **Chapter 3** works out M2’s binary case in the subcritical regime: the constant $A(p)$, quasi-stationary conditioning, and the Koenigs-function identity.
- **Chapters 4–5** solve M6 exactly (Riccati roots, geometric load, quasi-stationarity, burst statistics) and add M14 and M15 as the two single-cell comparators that organise the bursting/budding contrast.
- **Chapter 6** lifts M6 to the population layer: M16 is the standard comparator, M17 the burst-aware replacement (exact in the mean), M18 the branching question the intracellular model was built to answer, M19–M21 the spectrum around and beyond the catastrophe process.
- **Chapter 7** generalises M6’s catastrophe to two types with type-specific rates (M22), solved by hypergeometric reduction.
- **Chapter 8** shows why a release schedule alone cannot matter (M14 as negative result) and introduces the two nonlinearities that make it matter: intracellular depletion (M23) and extracellular fixed removal (M24).
- **Chapter 9** bends the rates: M8’s full law, the competitive chains M25–M26, the potential-driven cycles M27–M28, and the two specification-only feedback models M29–M30.
- **Chapter 10** closes with the coarsest model of the thesis (M31) and records, in its contributions table, which of these models are specifications or simulations that outrun the machinery (M26, M29’s variant, parts of M20–M21 and M24).